HOME TASKS-II

Task 1:(Easy-level)

Given a table of candidates and their skills, you're tasked with finding the candidates best suited for an open Data Science job. You want to find candidates who are proficient in Python, Tableau, and PostgreSQL.

Write a query to list the candidates who possess all of the required skills for the job. Sort the output by candidate ID in ascending order.

Assumption:

- There are no duplicates in the **candidates** table.

`candidates` **Table:**

Column Name	Type
candidate_id	integer
skill	varchar

`candidates` **Example Input:**

candidate_id	skill
123	Python
123	Tableau
123	PostgreSQL
234	R
234	PowerBI
234	SQL Server
345	Python
345	Tableau

Solution: select candidate_id from candidates

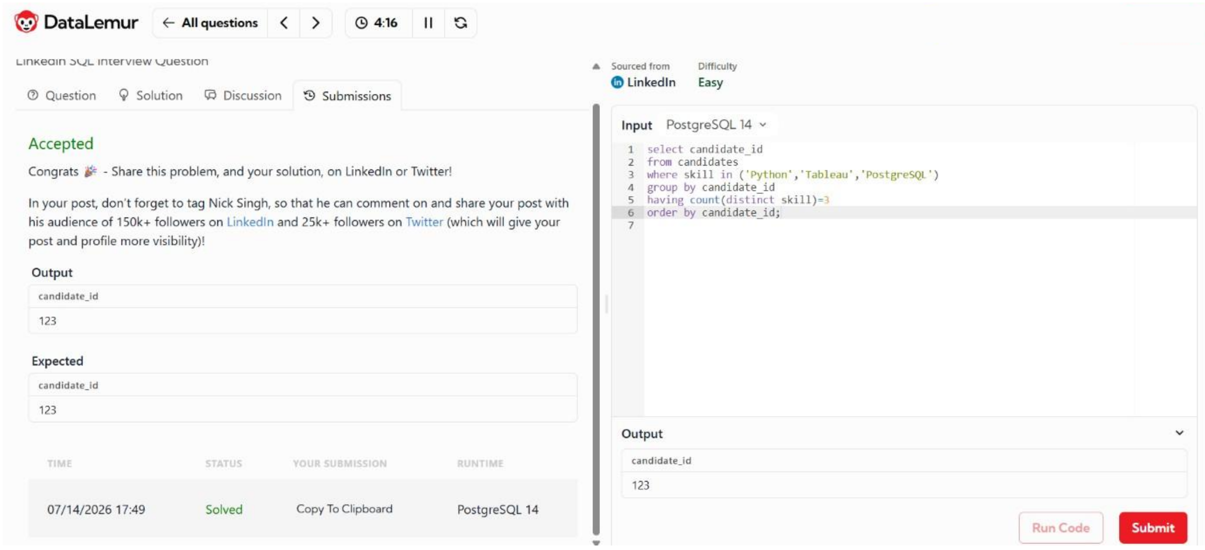
where skill in

('Python','Tableau','PostgreSQL') group by

candidate_id

having count(distinct skill)=3

order by candidate_id;



Task-2: Assume you're given two tables containing data about Facebook Pages and their respective likes (as in "Like a Facebook Page").

Write a query to return the IDs of the Facebook pages that have zero likes. The output should be sorted in ascending order based on the page IDs.

pages Table:

Column Name	Type
page_id	integer
page_name	varchar

pages Example Input:

page_id	page_name
20001	SQL Solutions
20045	Brain Exercises
20701	Tips for Data Analysts

page_likes Table:

Column Name	Type
user_id	integer
page_id	integer
liked_date	datetime

page_likes Example Input:

user_id	page_id	liked_date
111	20001	04/08/2022 00:00:00
121	20045	03/12/2022 00:00:00
156	20001	07/25/2022 00:00:00

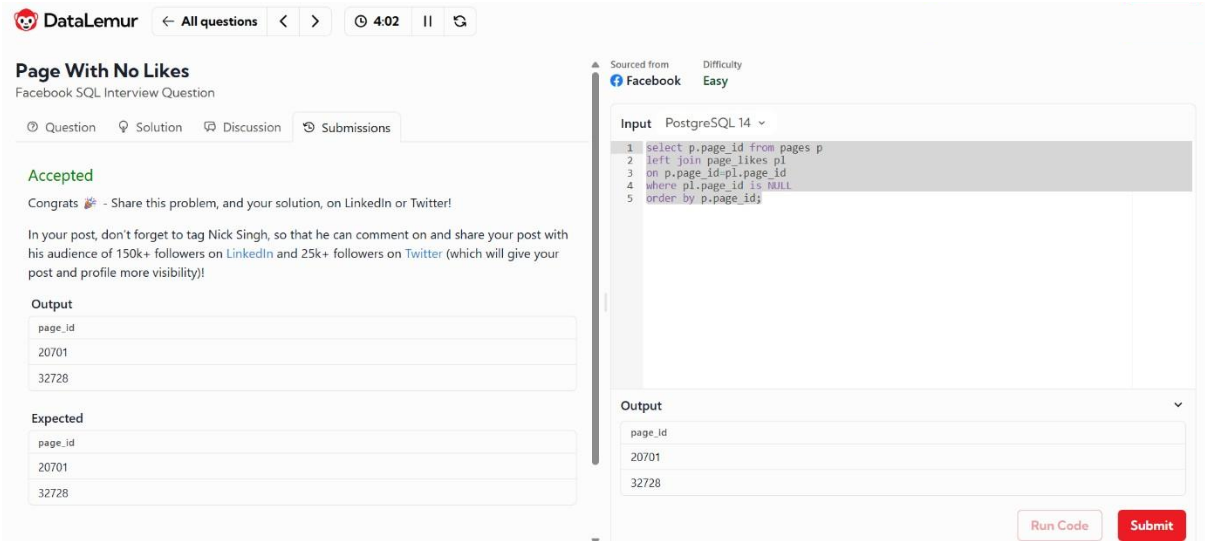
Solution: select p.page_id from pages p left

join page_likes pl

on p.page_id=pl.page_id where

pl.page_id is NULL

order by p.page_id;



Task-3: Tesla is investigating production bottlenecks and they need your help to extract the relevant data. Write a query to determine which parts have begun the assembly process but are not yet finished.

Assumptions:

- **parts_assembly** table contains all parts currently in production, each at varying stages of the assembly process.
- An unfinished part is one that lacks a **finish_date**.

`parts_assembly` Table

Column Name	Type
part	string
finish_date	datetime
assembly_step	integer

`parts_assembly` Example Input

part	finish_date	assembly_step
battery	01/22/2022 00:00:00	1
battery	02/22/2022 00:00:00	2
battery	03/22/2022 00:00:00	3
bumper	01/22/2022 00:00:00	1
bumper	02/22/2022 00:00:00	2
bumper		3
bumper		4

Solution: select part,assembly_step from parts_assembly

where finish_date is null;